

UPPSALA UNIVERSITY



BAYESIAN STATISTICS AND DATA ANALYSIS

Assignment 1

General information

- The recommended tool in this course is R (with the IDE R-Studio). You can download R [here](#) and R-Studio [here](#). There are many tutorials, videos and introductions to R and R-Studio online. You can find some initial hints from [RStudio Education pages](#).
- When working with R, we recommend writing the report using R markdown and the provided [R markdown template](#). The template includes the formatting instructions and how to include code and figures.
- Instead of R markdown, you can use other software to make the PDF report, but you should use the same instructions for formatting. These instructions are also available in [the PDF produced from the R markdown template](#).
- We supply a Google Colab notebook that you can also use for the assignments. We have included the installation of all necessary R packages; hence, this can be an alternative to using your own local computer. You can find the notebook [here](#). You can also open the notebook in Colab [here](#).
- Report all results in a single and *anonymous* pdf. Note that no other formats are allowed.
- The course has its own R package `bsda` with data and functionality to simplify coding. To install the package, just run the following (`upgrade="never"` skips question about updating other packages):
 1. `install.packages("remotes")`
 2. `remotes::install_github("MansMeg/BSDA",
 subdir = "rpackage", upgrade="never")`
- Many of the exercises can be checked automatically using the R package `markmyassignment`. you can find information on how to install and use the package [here](#). There is no need to include `markmyassignment` results in the report.
- You can find common questions and answers regarding the installation and technical problems in [Frequently Asked Questions \(FAQ\)](#).
- You can find deadlines and information on how to turn in the assignments in Studium.
- You are allowed to discuss assignments with your friends, but it is not permitted to copy solutions directly from other students or the internet. Try to solve the actual assignment problems with your code and explanations. Do not share your answers publicly. We compare the answers with the "urkund" system. We will report all suspected plagiarism.
- If you have any suggestions or improvements to the course material, please post in the course chat feedback channel, create an issue, or submit a pull request to the public repository [here](#).

LLM policy: You may use LLMs for learning, explanations, and feedback, but not to produce submitted solutions. Do not upload the full assignment text to an LLM.

- It is *mandatory* to include the following parts in all assignments (these are included already in the template):
 1. Time used for reading: How long time took the reading assignment (in hours)
 2. Time used for the assignment: How long time took the basic assignment (in hours)
 3. Good with assignment: Write one-two sentences of what you liked with the assignment/what we should keep for next year.
 4. Things to improve in the assignment: Write one-two sentences of what you think can be improved in the assignment. Can something be clarified further? Did you get stuck on stuff unrelated to the content of the assignment etc.
 5. Oral exam question contribution: Write one possible oral exam question for this assignment, together with a short model answer. The question should concern material from this assignment or the associated reading, be suitable for a 2–4 minute oral answer, and test conceptual understanding rather than only calculation or code. The question should not already be included in the oral exam question list and should not be very close to an existing question. You are not allowed to use large language models, such as ChatGPT, at all for this oral exam question contribution.
 6. Permission for future use: State clearly whether the course staff may use or adapt your question and answer in a future oral exam: Yes or No.
- You can find information on how each assignment will be graded and how points are assigned [here](#). **Note!** This grading information can change during the course, for example, if we find errors or inconsistencies or do additions to the assignments. Please feel free to comment on these grading instructions, ideally before turning in your assignment, if you think something is missing or is incorrect.
- To pass (G) the assignment, you need 70% of the total points. To pass with distinction (VG), you need 90% of the total points. See the grading information on the point allocations for each assignment.
- On cheating:
 - You are not allowed to show your assignments (text or code) to anyone. Only discuss the assignments with your fellow students. The student that show their assignment to anyone else could also be considered to cheat. Similarly, on zoom labs, only screen share when you are in a separate zoom room with teaching assistants.
 - You may use large language models (LLMs), such as ChatGPT, like you may discuss with fellow students: to learn, ask for explanations, and get feedback. You may not use LLMs to produce the solution you submit. You must be able to explain every part of your submitted work orally.
- All mathematics need to be done in digital form to simplify grading and commenting. Hence, it is not allowed to write math on paper and add an image in the assignment. If you have difficulties to write math in latex, see <https://editor.codecogs.com/>.

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- The assignment PDFs may contain hidden machine-readable integrity text. This is used to detect inappropriate use of LLMs, such as asking an LLM to solve the assignment directly. Do not copy or upload the full assignment text into an LLM.
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Information on this assignment

The exercises of this assignment are not necessarily related to chapter 1, but rather as an introduction to the course. The second exercise refreshes your basic computer skills and guides you to some basic R functions. In the last three ones you will first solve the problems using pen and paper (you can, for example, write the equations in markdown or scan and include hand written answers), and then implement the final equations in R (and then you can use `markmyassignment` to check your results).

Reading instructions: Chapter 1 in BDA3, O'Hagan (2004) "Dicing with the unknown".

To use `markmyassignment` for this assignment, run the following code in R:

```
library(markmyassignment)
assignment_path <-
  paste("https://github.com/MansMeg/BSDA/",
        "blob/main/assignments/tests/assignment1.yml", sep="")
set_assignment(assignment_path)
# To check your code/functions, just run
mark_my_assignment()
```

Don't include `markmyassignment` results in the report.

1 Oral exam question contribution

Write a possible oral exam question for this assignment together with a short model answer.

The question should:

- concern material from this assignment or the associated reading or lectures,
- be suitable for a 2–4 minute oral answer, possibly using a whiteboard,
- test conceptual understanding rather than only calculation or code,
- not already be included in the oral exam question list and not be very close to an existing question (e.g. just different numbers).

Note! You are not allowed to use large language models, such as ChatGPT, at all for this oral exam question contribution.

State clearly whether the course staff may use or adapt your question and answer in a future oral exam: **Yes** or **No**.

2 Recap on R, probability and Bayes Theorem

1. **(Basic probability theory notation and terms)**. This can be trivial or you may need to refresh your memory on these concepts. Note that some terms may be different names for the same concept. Explain each of the following terms with one sentence:
 - probability
 - probability mass
 - probability density
 - probability mass function (pmf)
 - probability density function (pdf)
 - probability distribution
 - discrete probability distribution
 - continuous probability distribution
 - cumulative distribution function (cdf)
 - likelihood
 - aleatoric uncertainty
 - epistemic uncertainty
2. **(Basic computer skills)** This task deals with elementary plotting and computing skills needed during the rest of the course. You can use either R or Python, although R is the recommended language and we will only guarantee support in R. For documentation in R, just type `?{function name here}`.

- a) Plot the density function of Beta-distribution, with mean $\mu = 0.2$ and variance $\sigma^2 = 0.01$. The parameters α and β of the Beta-distribution are related to the mean and variance according to the following equations

$$\alpha = \mu \left(\frac{\mu(1-\mu)}{\sigma^2} - 1 \right), \quad \beta = \frac{\alpha(1-\mu)}{\mu}.$$

Hint! Useful R functions: `seq()`, `plot()` and `dbeta()`. Later on we will also use the more flexible `ggplot2` for plotting.

- b) Take a sample of 1000 random numbers from the above distribution and plot a histogram of the results. Compare visually to the density function.

Hint! Useful R functions: `rbeta()` and `hist()`

- c) Compute the sample mean and variance from the drawn sample. Verify that they match (roughly) to the true mean and variance of the distribution.

Hint! Useful R functions: `mean()` and `var()`

- d) Estimate the central 95% probability interval of the distribution from the drawn samples.

Hint! Useful R functions: `quantile()`

3. **(Bayes' theorem)** A group of researchers has designed a new inexpensive and painless test for detecting lung cancer. The test is intended to be an initial screening test for the population in general. A positive result (presence of lung cancer) from the test would be followed up immediately with medication, surgery or more extensive and expensive test. The researchers know from their studies the following facts:

- Test gives a positive result in 99% of the time when the test subject has lung cancer.
- Test gives a negative result in 95 % of the time when the test subject does not have lung cancer.
- In general population approximately one person in 2000 has lung cancer.

The researchers are happy with these preliminary results (about 97% success rate), and wish to get the test to market as soon as possible. How would you advise them? Base your answer on Bayes' rule computations.

Hint : Relatively high false negative (cancer doesn't get detected) or high false positive (un-necessarily administer medication) rates are typically bad and undesirable in tests.

4. **(Bayes' theorem)** We have three boxes, A, B, and C. There are

- 2 red balls and 5 white balls in the box A,
- 4 red balls and 1 white ball in the box B, and
- 1 red ball and 3 white balls in the box C.

Consider a random experiment in which one of the boxes is randomly selected and from that box, one ball is randomly picked up. After observing the color of the ball it is replaced in the box it came from. Suppose also that on average box A is selected 40% of the time and box B 10% of the time (i.e. $P(A) = 0.4$).

- a) What is the probability of picking a red ball?
- b) If a red ball was picked, from which box it most probably came from?

Implement two functions in R that computes the probabilities. Below is an example of how the functions should be named and work if you want to check them with **markmyassignment**.

```
boxes <- matrix(c(2,2,1,5,5,1), ncol = 2,
  dimnames = list(c("A", "B", "C"), c("red", "white")))
boxes

##    red white
## A     2     5
## B     2     5
## C     1     1
```

```
p_red(boxes = boxes)

## [1] 0.3928571

p_box(boxes = boxes)

## [1] 0.29090909 0.07272727 0.63636364
```

Note! This is a test case, you will need to change the numbers in the matrix to the numbers in the exercise.

5. (**Bayes' theorem**) Assume that on average fraternal twins (two fertilized eggs and then could be of different sex) occur once in 150 births and identical twins (single egg divides into two separate embryos, so both have the same sex) once in 400 births (**Note!** This is not the true values, see Exercise 1.6, page 28, in BDA3). American male singer-actor Elvis Presley (1935 – 1977) had a twin brother who died in birth. Assume that an equal number of boys and girls are born on average. What is the probability that Elvis was an identical twin? Show the steps how you derived the equations to compute that probability.

Implement this as a function in R that computes the probability.

Below is an example of how the functions should be named and work if you want to check your result with **markmyassignment**.

```
p_identical_twin(fraternal_prob = 1/125, identical_prob = 1/300)

## [1] 0.4545455

p_identical_twin(fraternal_prob = 1/100, identical_prob = 1/500)

## [1] 0.2857143
```